

AZ-900

Learning path 2: Azure architecture and services



Agenda



Learning path 1: Cloud concepts

Learning path 2: Azure architecture and services

Learning path 3: Azure management and governance

Describe the core architectural components of Azure



What is Microsoft Azure?

Microsoft Azure Service Categories

Compute

Virtual Machines
App Service
Azure Functions
Container Apps
Azure Kubernetes Service

Networking

Virtual Network
Load Balancer
VPN Gateway
ExpressRoute
Azure DNS

Storage

Blob Storage
Azure Files
Queue Storage
Table Storage
Managed Disks

Databases

Azure SQL
Azure Cosmos DB
PostgreSQL Flex
MySQL Flex
Redis Cache

AI + ML

Azure OpenAI
AI Services
Machine Learning
Bot Service
AI Search

Identity + Security

Microsoft Entra ID
Key Vault
Defender for Cloud
DDoS Protection
Firewall

DevOps + Management

Azure DevOps
Azure Monitor
ARM Templates
Azure Policy
Cost Management

IoT

IoT Hub
IoT Central
Digital Twins
Stream Analytics
Event Hubs

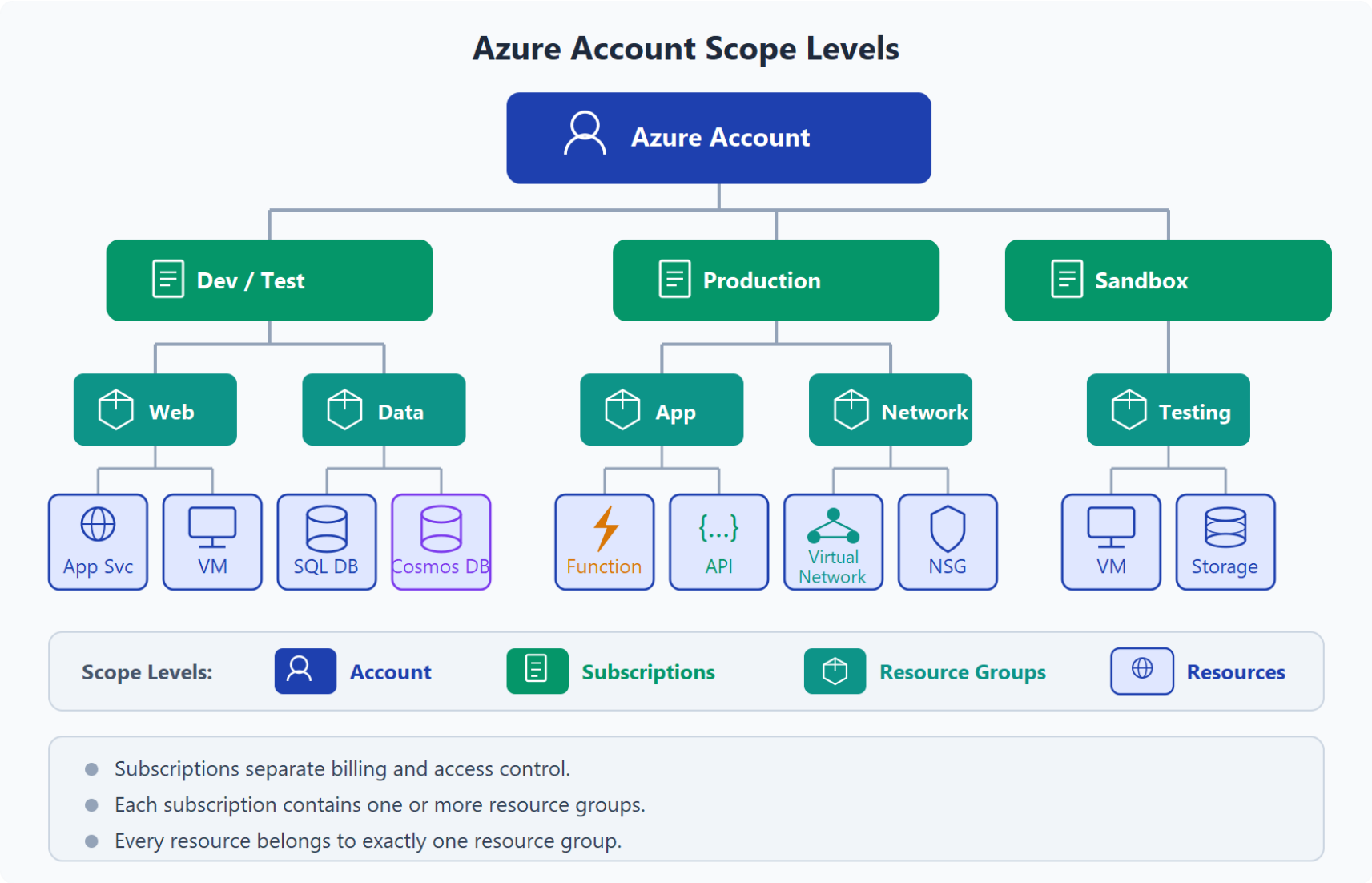
Analytics

Synapse Analytics
Data Factory
Databricks
HDInsight
Azure Data Explorer

Integration

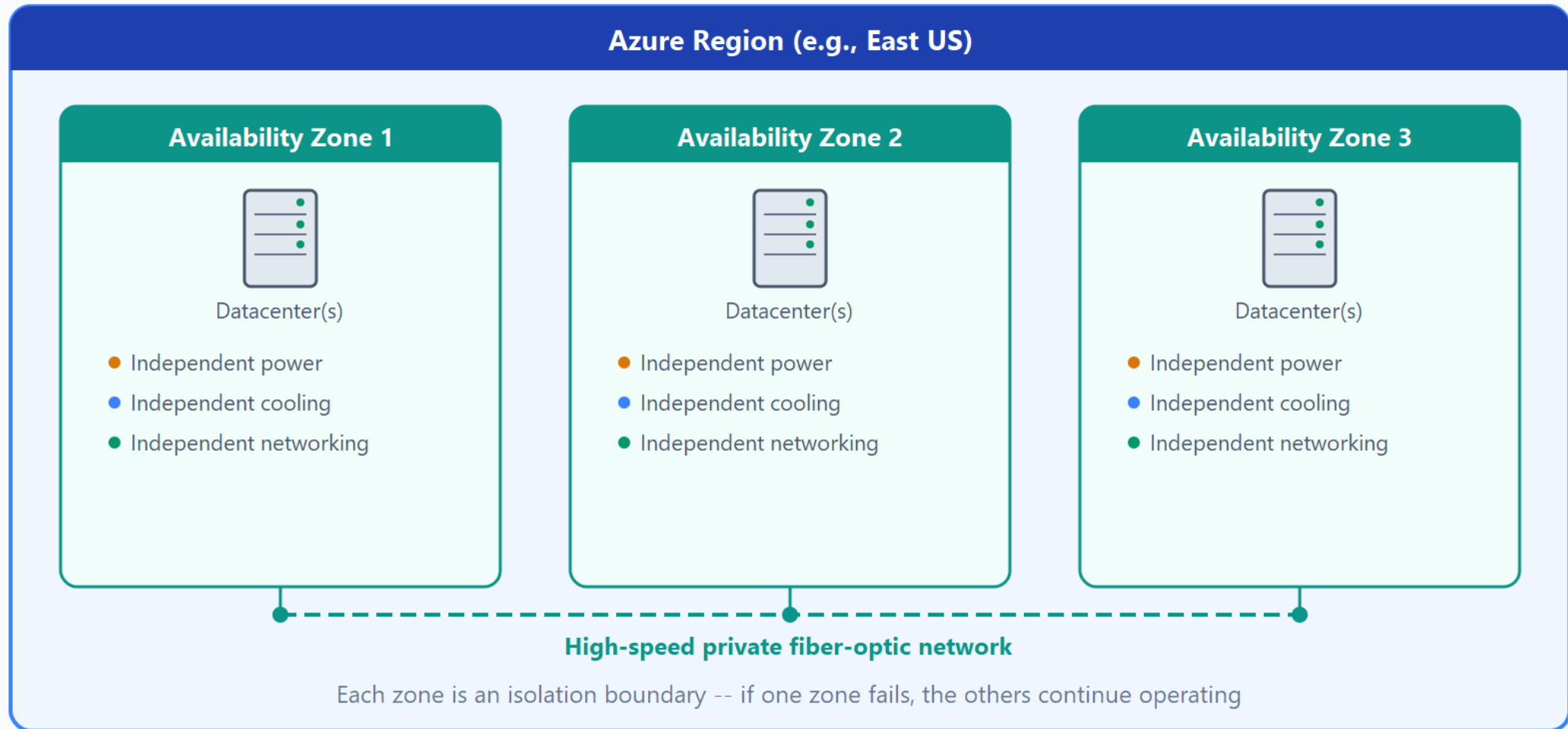
Logic Apps
API Management
Service Bus
Event Grid
Notification Hubs

Get started with Azure accounts

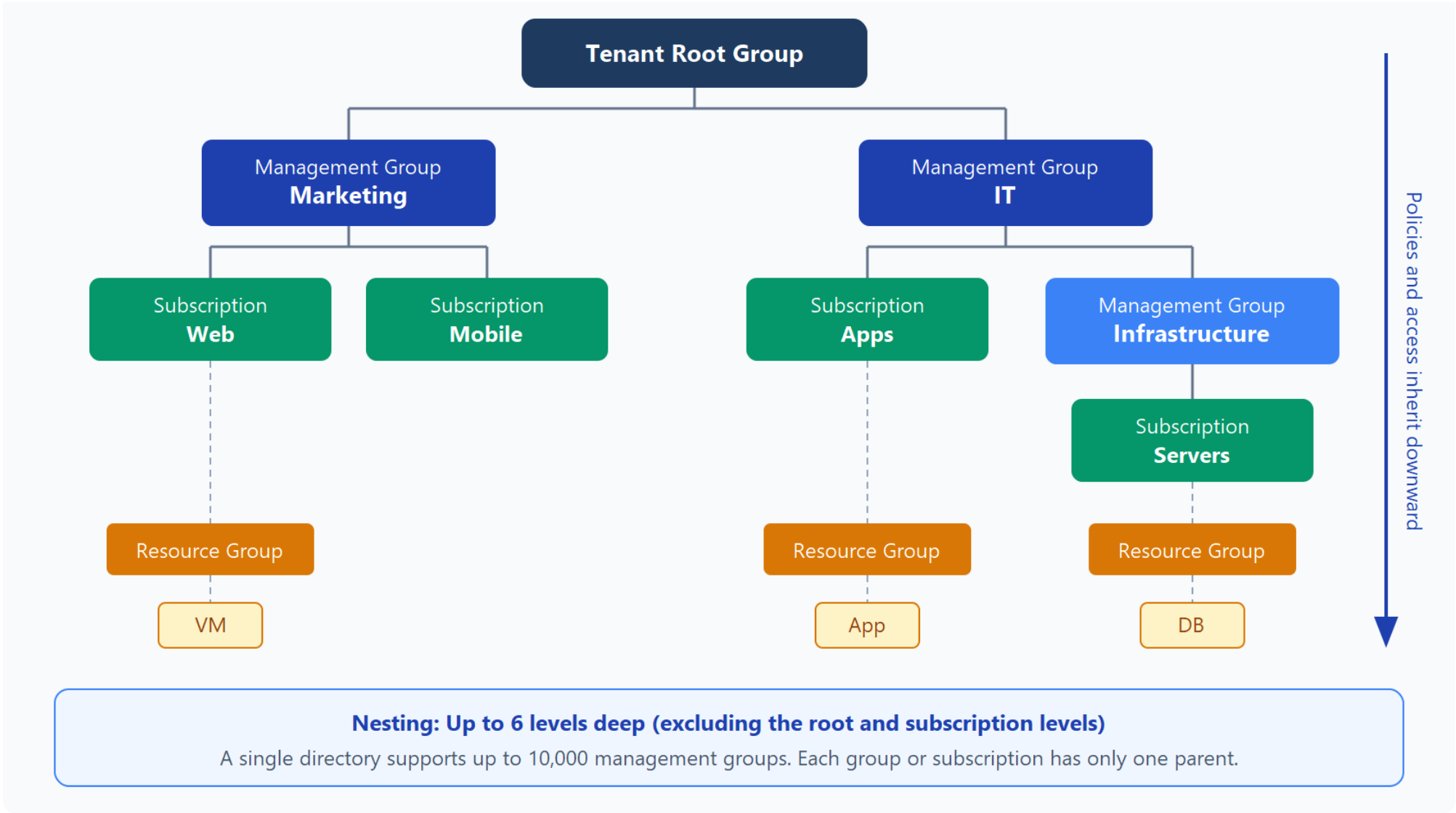


Azure physical infrastructure

Availability Zones in an Azure Region



Azure management infrastructure



Knowledge check



1 Which statement is true about Azure regions and availability zones?

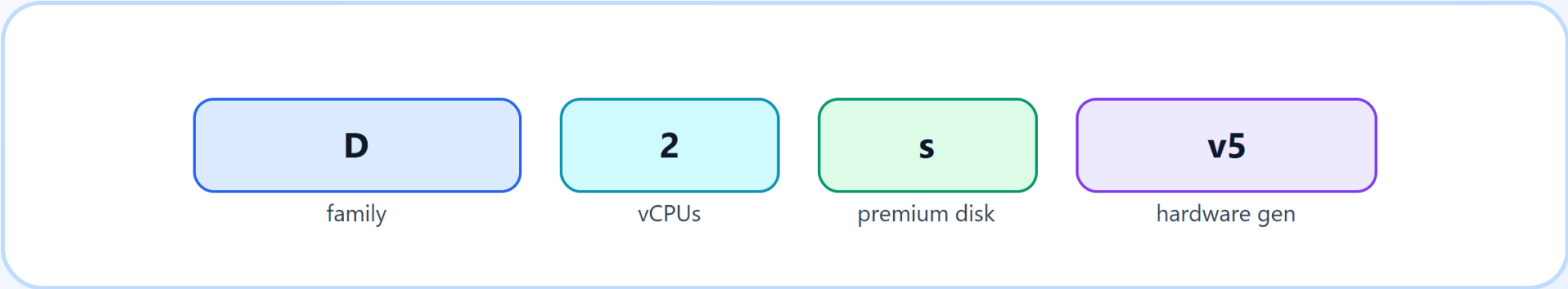
- ☐ Every region has exactly 3 zones
- ☐ Availability zones are separate datacenters within a region
- ☐ Regions are only logical groupings, not physical

Describe Azure compute services



Decoding Azure Virtual Machine names

How to read Standard_D2s_v5



Azure Virtual Machines families

VM size families

Choose the family that best matches workload behavior.

B

burstable

D

general

E

memory

F

compute

M

large memory

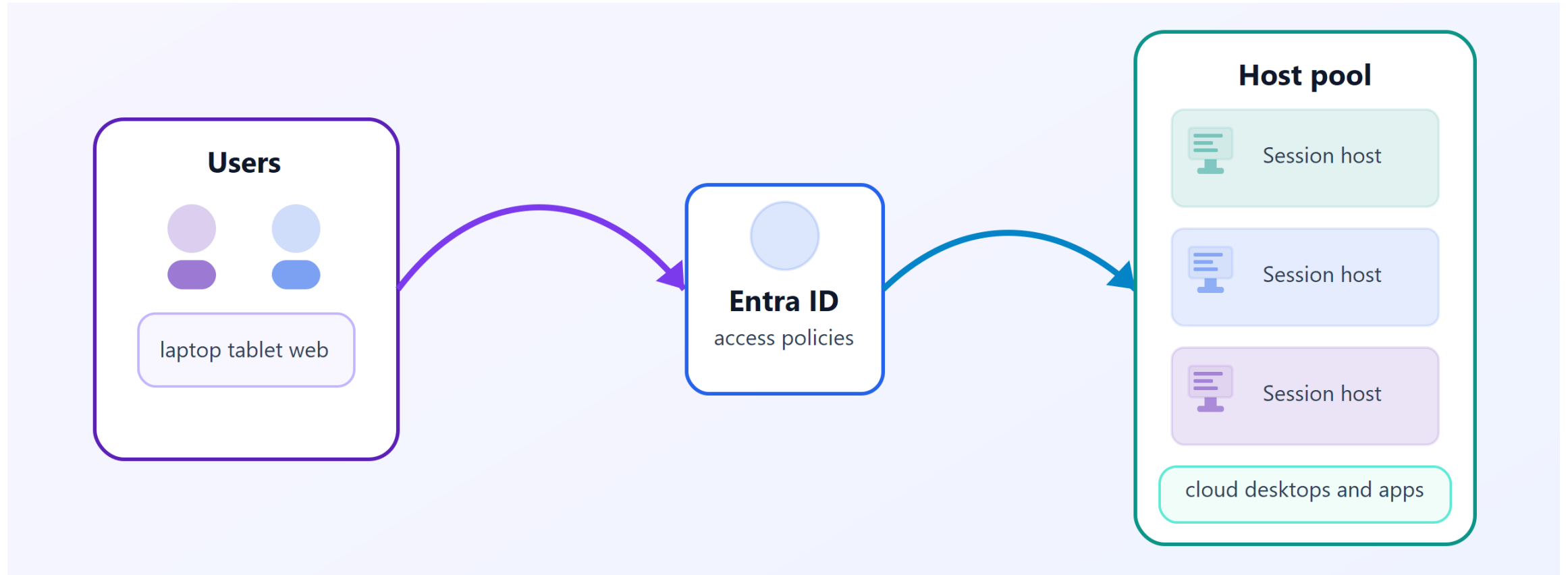
L

storage

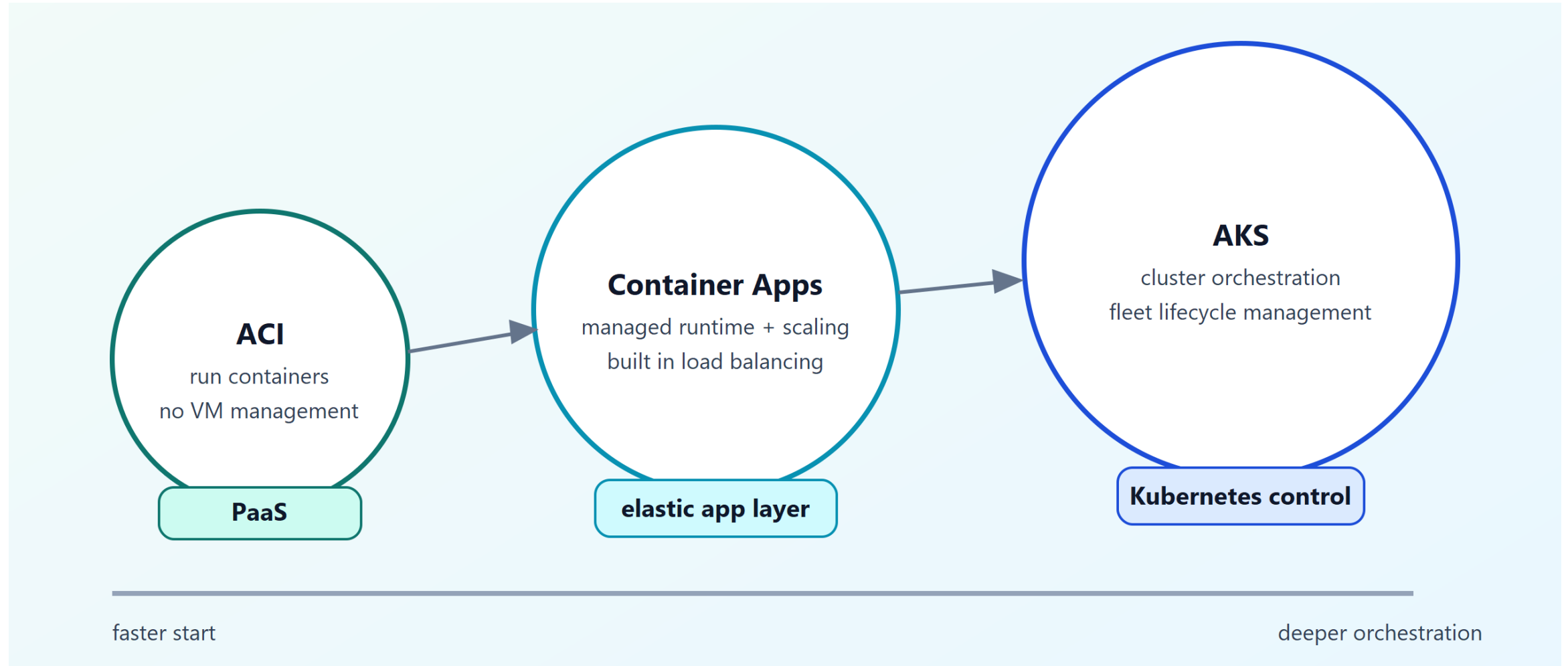
N

GPU

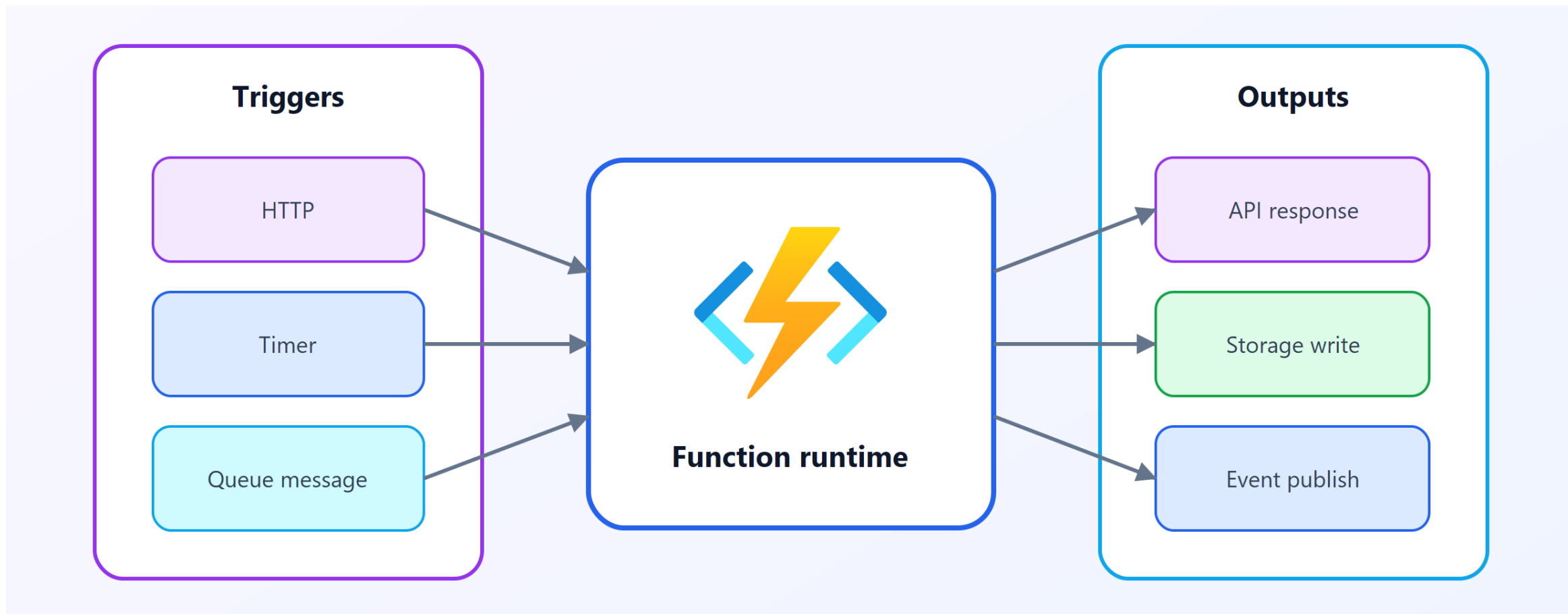
Azure Virtual Desktop



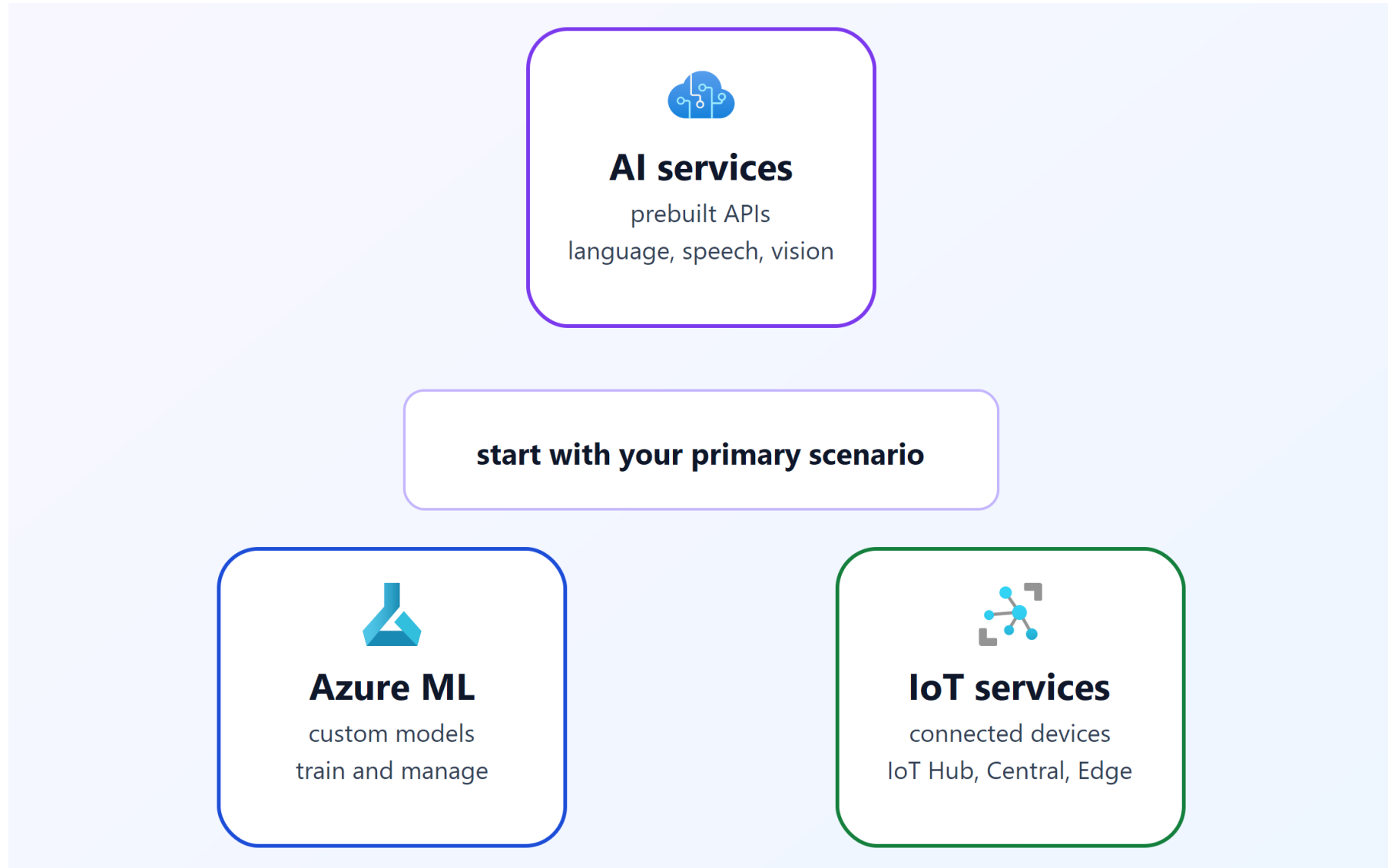
Azure container services



Azure Functions



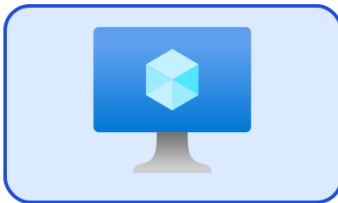
AI, machine learning, and IoT and Edge services



Application hosting options

Virtual Machines

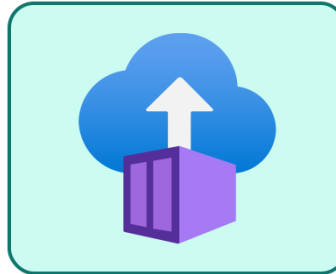
OS and config control
custom hosting setup
you patch and maintain



more control

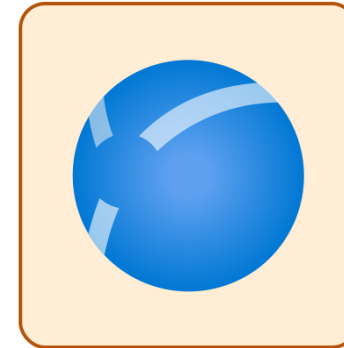
Containers

component isolation
portable deployments
independent scaling



App Service

managed platform
autoscale and HA
integrated deployment



less operational effort

Knowledge check



1 Which Azure service is best when you want to run containerized applications without managing VM infrastructure?

- ☐ Azure Virtual Machines
- ☐ Azure Functions
- ☐ Azure Kubernetes Service (AKS)

Describe Azure networking services



Azure virtual networks

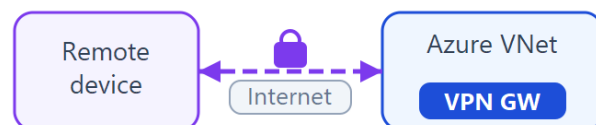
- VNets provide isolation, segmentation, internet and cross-network connectivity
- Public endpoints have public IPs; private endpoints use IPs within the VNet address space
- Network Security Groups (NSGs) filter traffic with inbound/outbound rules by IP, port, and protocol
- VNet peering connects VNets (even cross-region) over the Microsoft private backbone
- On-premises connectivity: point-to-site VPN, site-to-site VPN, and Azure ExpressRoute

Azure VPN Gateway

VPN gateway connectivity models

All traffic encrypted inside private tunnels across the public internet

Point-to-site



A single client device connects to an Azure VPN gateway over an encrypted tunnel.

Remote workers, mobile users

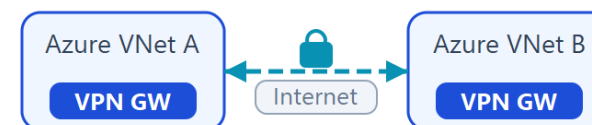
Site-to-site



An on-premises VPN device connects to an Azure VPN gateway through an IPsec/IKE tunnel.

Branch offices, datacenters

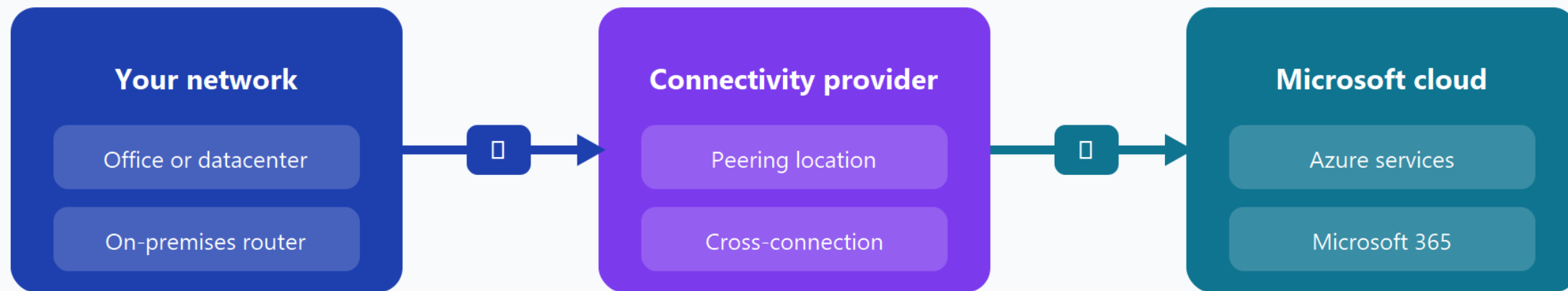
Network-to-network



Two Azure virtual networks connect their VPN gateways across regions through an encrypted tunnel.

Cross-region Azure connectivity

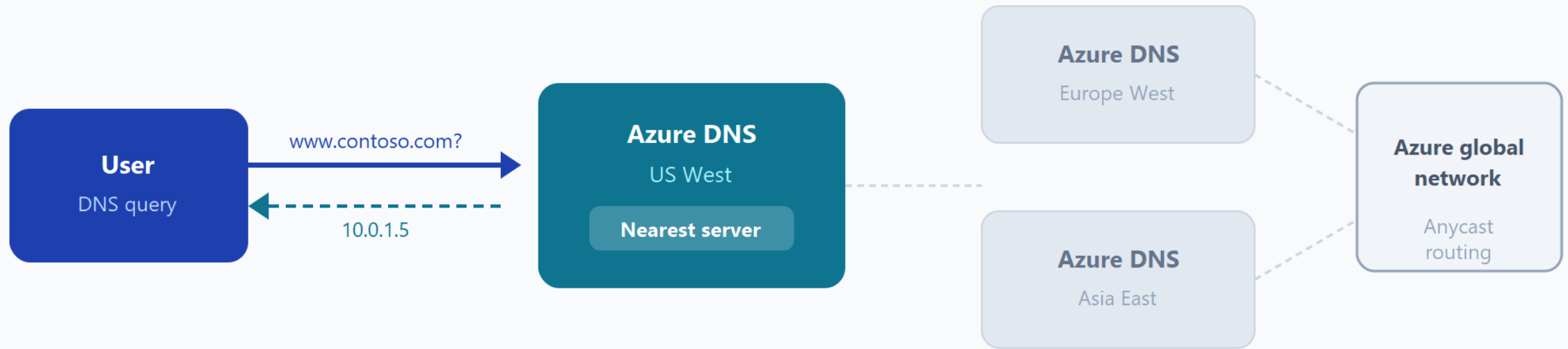
Azure ExpressRoute



ExpressRoute circuit — private connection

More reliable · Faster speeds · Consistent latency · Higher security

Azure DNS



Anycast routes each query to the closest healthy DNS server for fast, reliable resolution

Knowledge check



1 What is the primary purpose of a virtual network (VNet) in Azure?

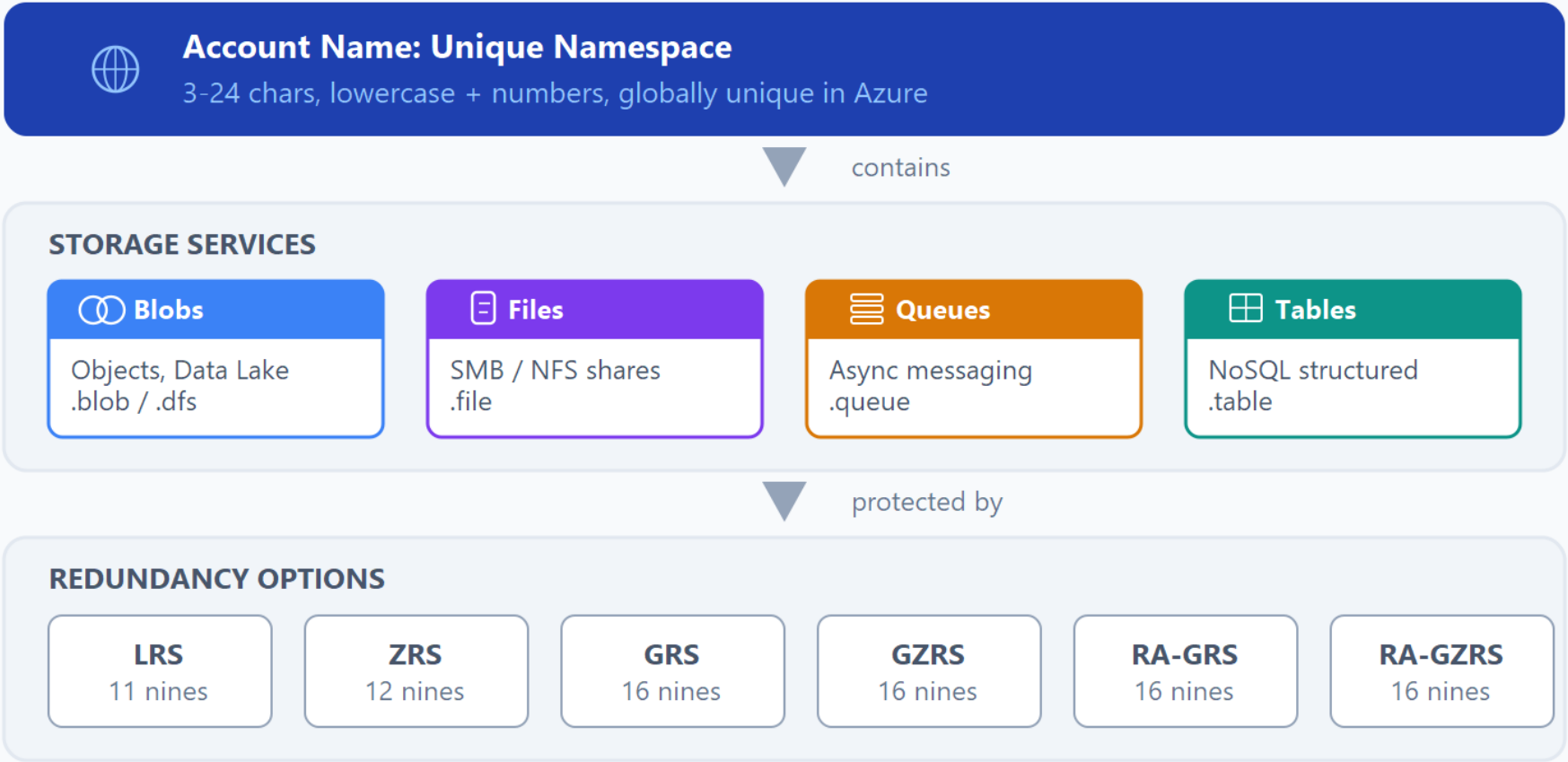
- ☐ Store unstructured data
- ☐ Provide isolated network communication for Azure resources
- ☐ Manage identities and access

Describe Azure storage services



Azure storage accounts

Anatomy of an Azure Storage Account



A storage account provides a unique namespace, contains one or more services, and is protected by a redundancy option.

Azure storage redundancy

- LRS replicates data 3x in a single datacenter (11 nines durability); lowest cost
- ZRS replicates across 3 availability zones in the primary region (12 nines durability)
- GRS uses LRS in both primary and secondary regions (16 nines durability)
- RA-GRS and RA-GZRS enable read access to secondary region data before failover

Azure storage services

Azure Storage Services

Azure Blobs

Object store for text and binary data
Includes Data Lake Storage Gen2

Azure Files

Managed file shares via SMB/NFS
Cloud and on-premises access

Azure Queues

Messaging store for async processing
Messages up to 64 KB each

Azure Disks

Block-level storage volumes for VMs
Fully managed by Azure

Azure Tables

NoSQL store for structured data
Non-relational data storage

Azure data migration options

Azure Migrate

Online

Real-time migration hub

Single portal to assess and migrate
on-premises infrastructure to Azure

Discovery and assessment
Server migration
Database migration
Web app migration
ISV tool integration

Azure Data Box

Physical

Offline bulk data transfer

Ship a secure storage device
with up to 80 TB capacity

One-time bulk migration
Periodic large uploads
Data export from Azure
NIST 800-88r1 disk wipe
End-to-end tracking in portal

Azure file movement options

Azure File Movement Tools

AzCopy

Command Line

Upload and download files
Copy between accounts
One-way synchronization
Cross-cloud provider

Sync Direction
One-way (source to destination)

Scripted / automated tasks

Storage Explorer

GUI App

Graphical interface
Windows, macOS, Linux
Uses AzCopy on backend
Manage blobs and files

Interaction Model
Manual upload / download

Visual browsing and management

Azure File Sync

Agent Service

Bi-directional sync
Cloud tiering
SMB, NFS, FTPS protocols
Multisite caching

Sync Direction
Bi-directional (automatic)

Hybrid file server scenarios

Each tool serves different file movement needs -- from scripted transfers to always-on hybrid sync

Knowledge check



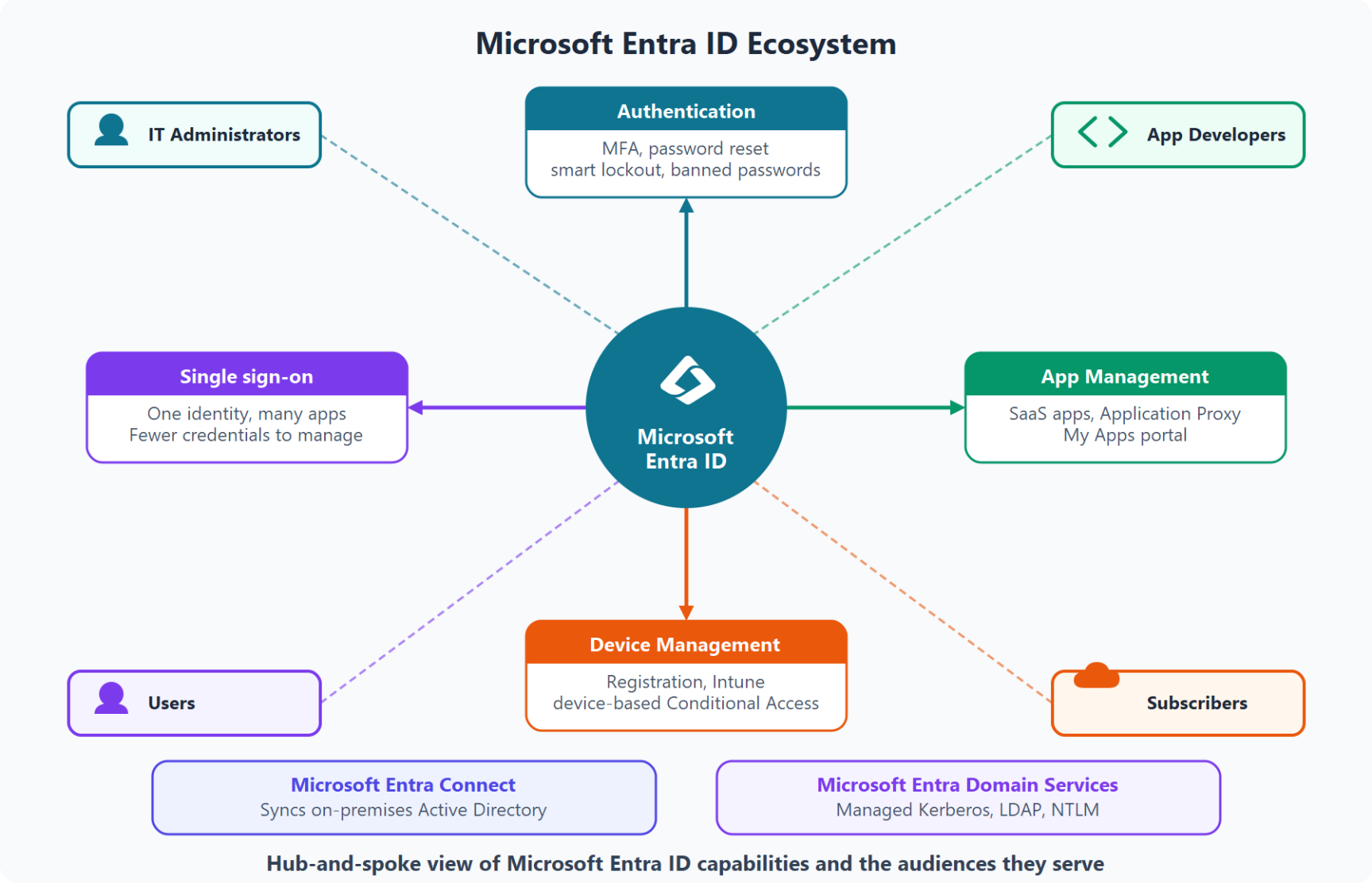
1 Which storage type is most appropriate for unstructured objects like images, backups, and logs?

- ☐ Azure Queue Storage
- ☐ Azure Files
- ☐ Azure Blob Storage

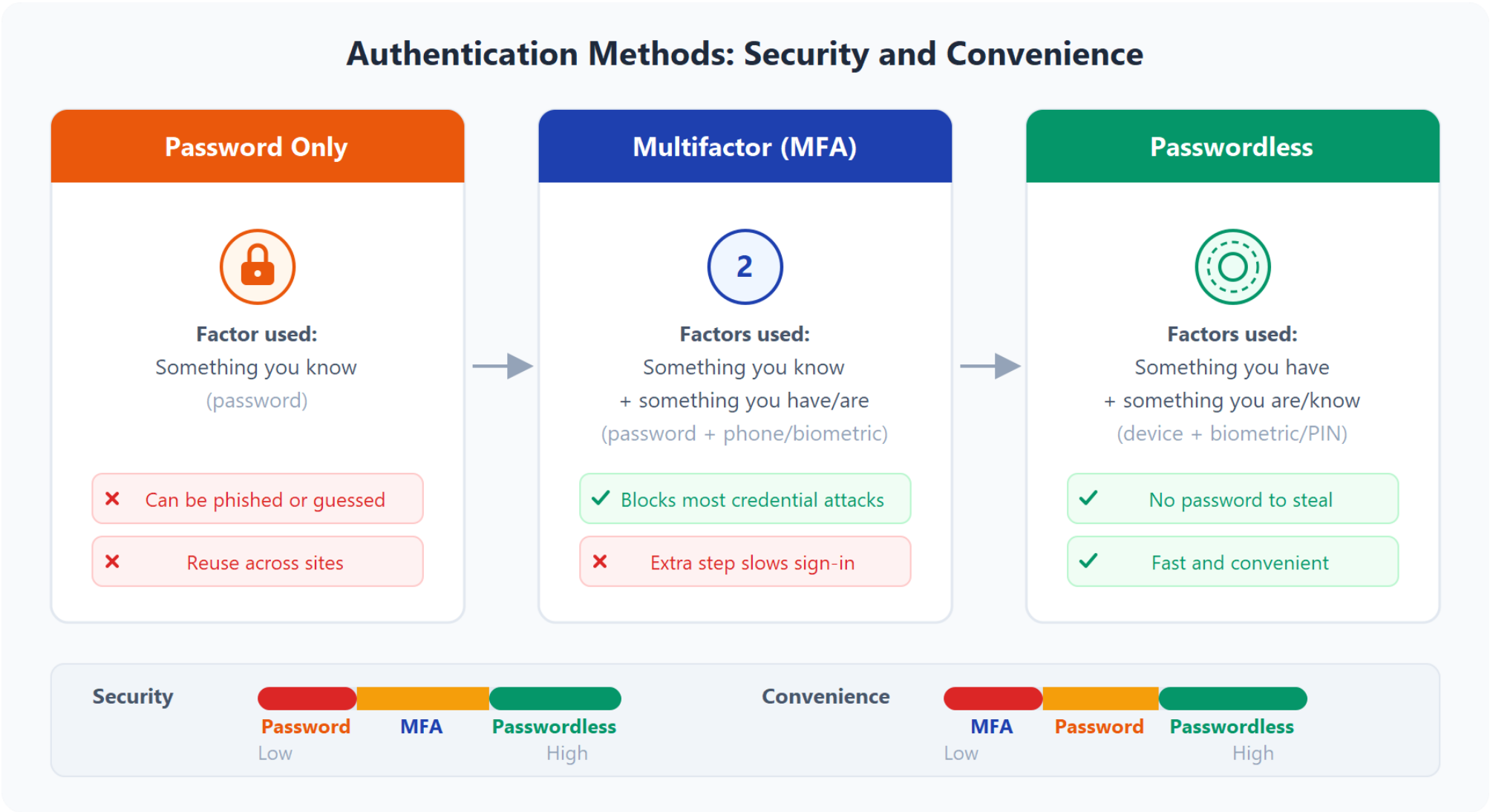
Describe Azure identity, access, and security



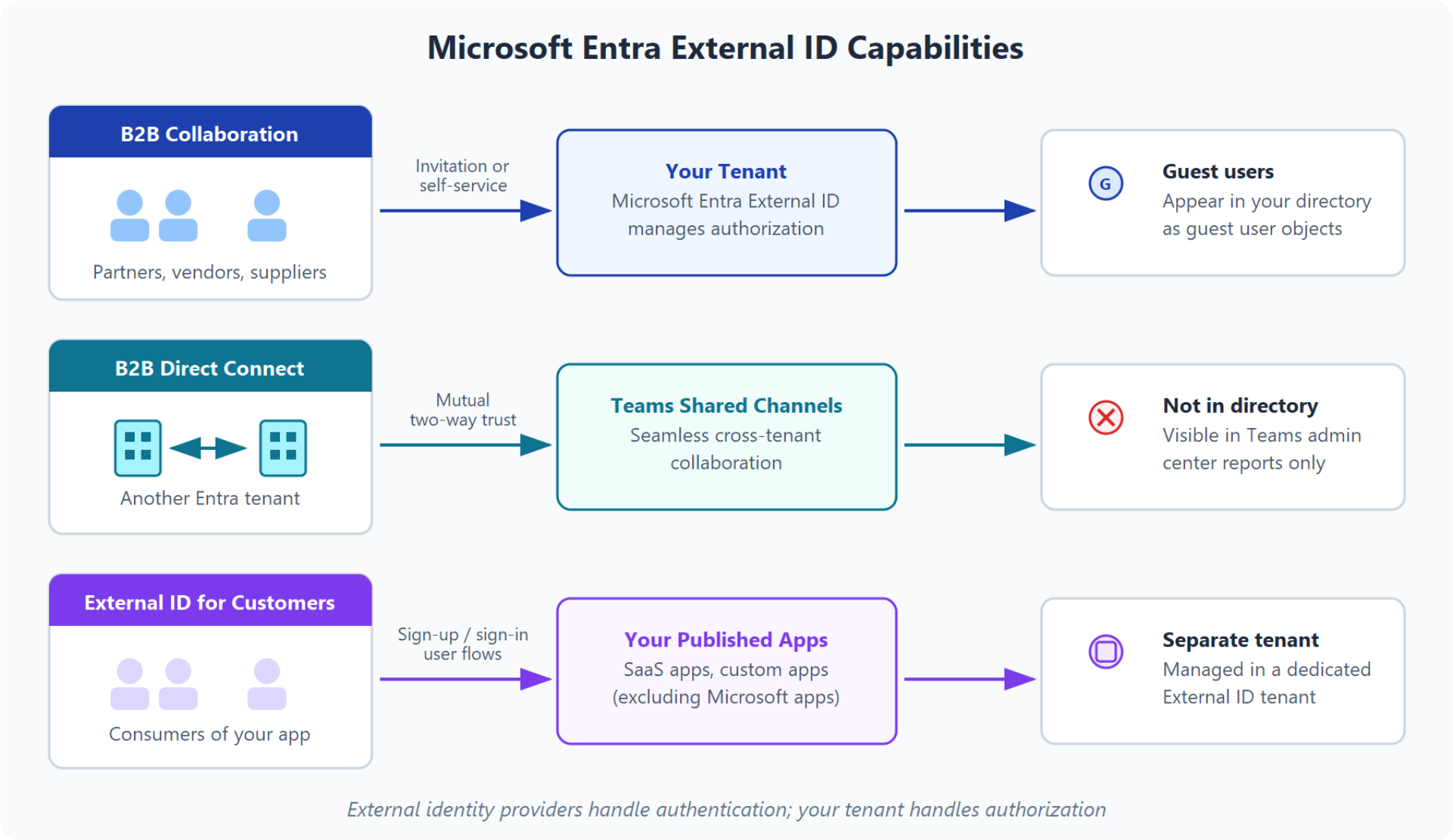
Microsoft Entra ID and Domain Services



Authentication methods

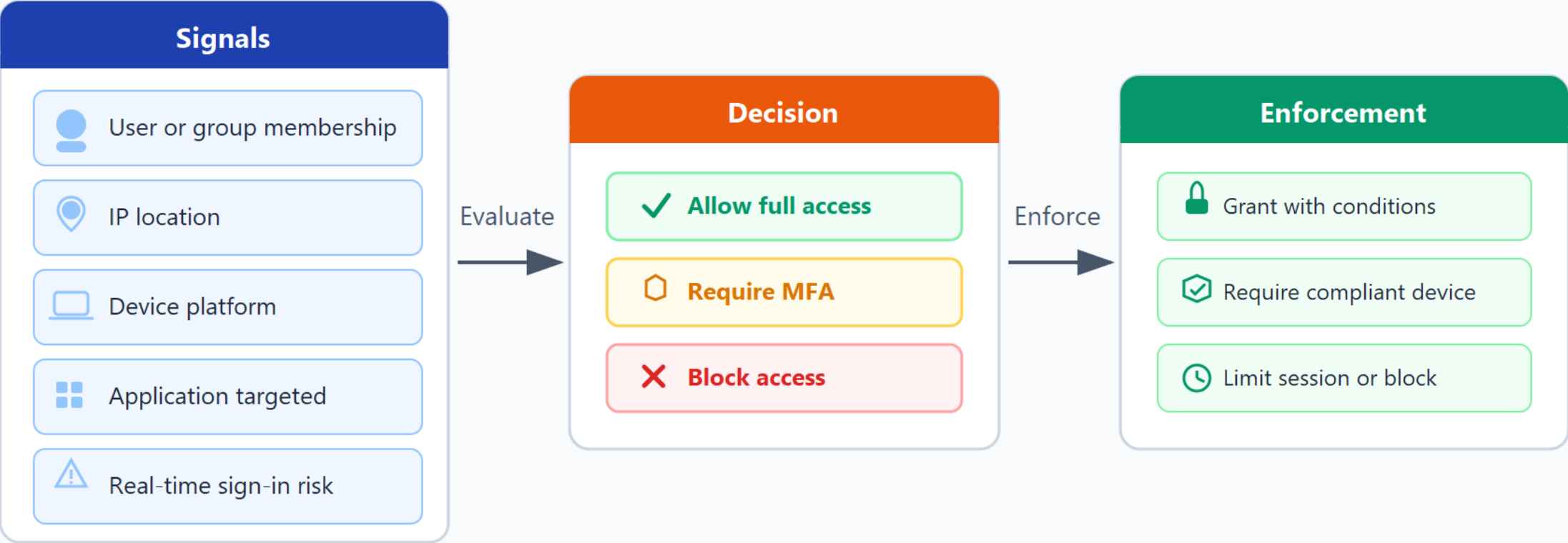


External identities



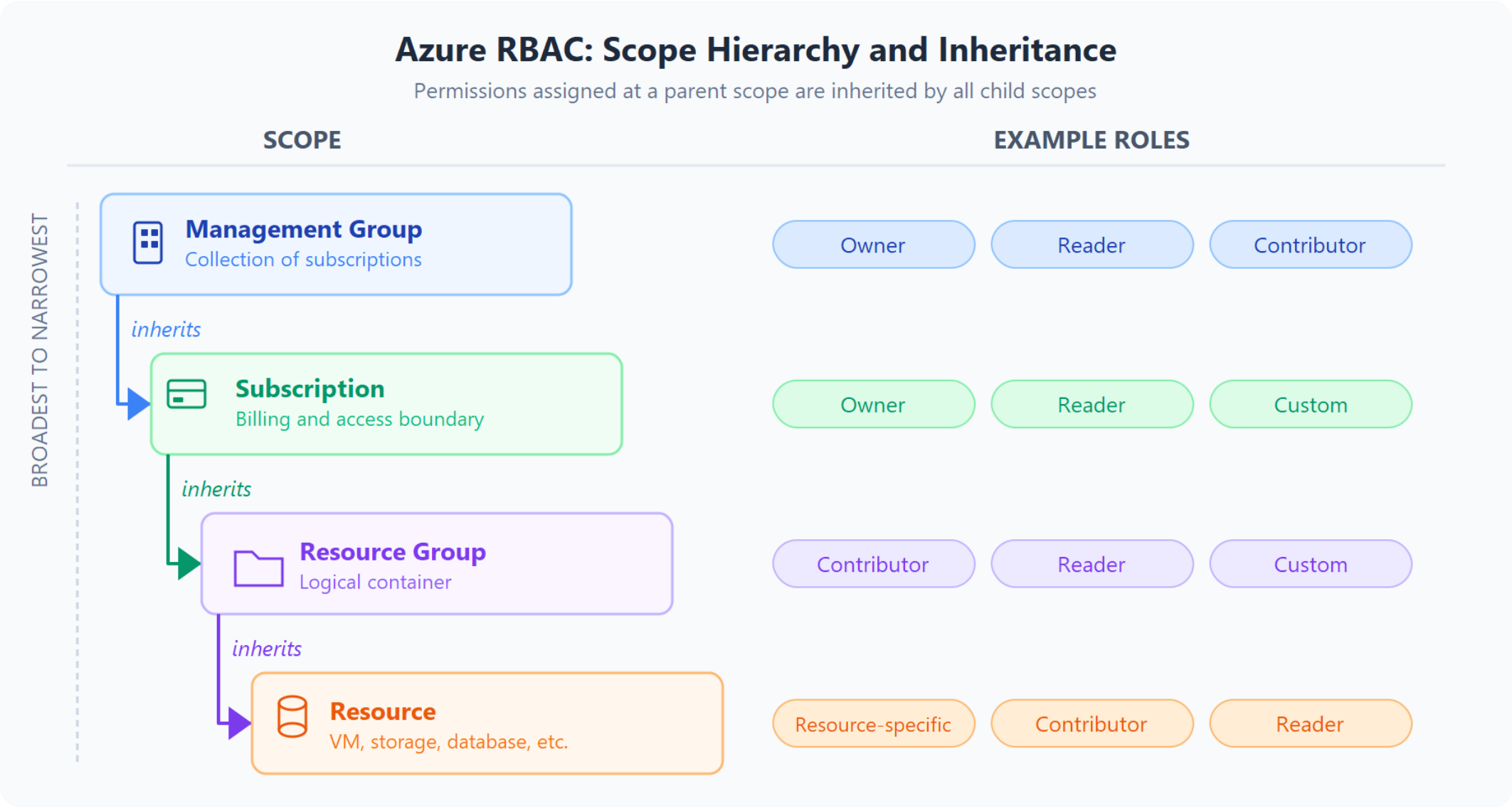
Conditional Access

Conditional Access: Signal to Enforcement



Conditional Access evaluates every sign-in against your policies in real time

Role-based access control (RBAC)



Zero Trust model

Zero Trust Guiding Principles

Verify explicitly



- Authenticate and authorize every request
- Evaluate all available data points
- Consider identity, location, device, service, and data classification

Least privilege access



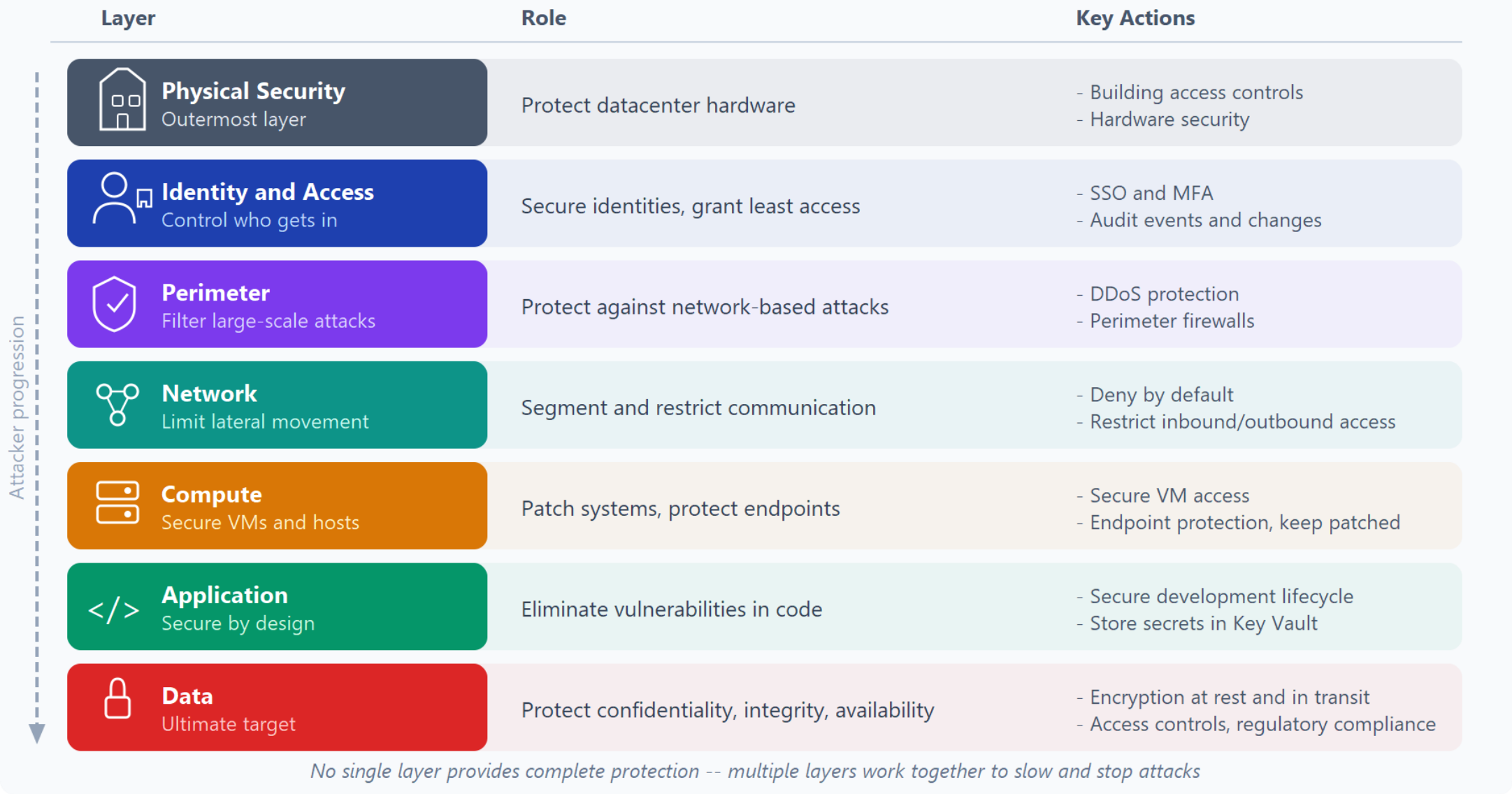
- Limit access with Just-In-Time and Just-Enough-Access (JIT/JEA)
- Apply risk-based adaptive policies
- Protect data with policy-driven access controls

Assume breach

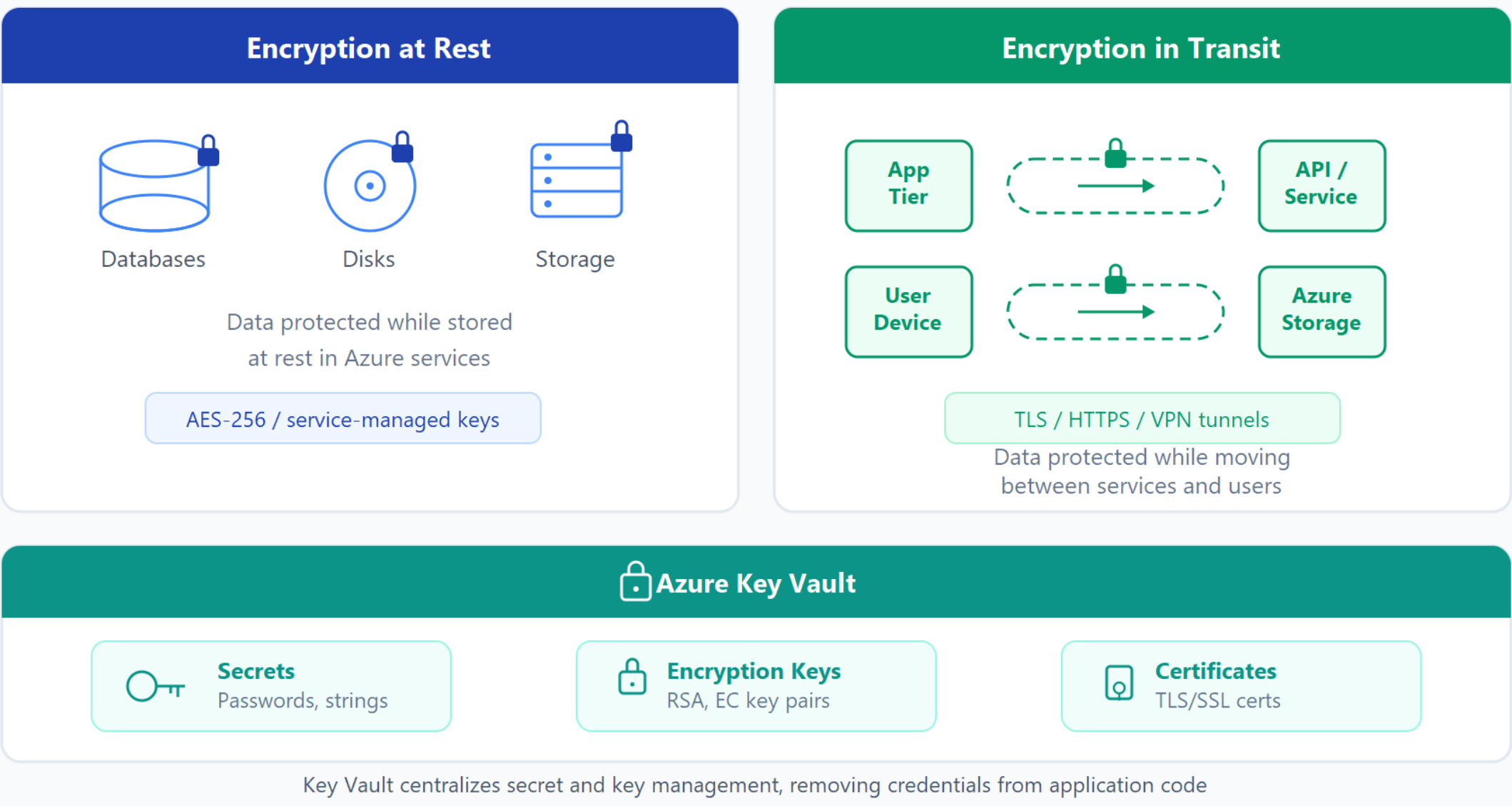


- Limit the potential impact and segment access
- Verify end-to-end encryption
- Use analytics for visibility, threat detection, and defense improvement

Defense in depth



Encryption and key management



Microsoft Defender for Cloud

Three pillars of cloud security management

 **Continuously Assess**

Know your security posture

Key Actions

- Identify and track vulnerabilities
- VM vulnerability scans
- Container registry assessments
- SQL server assessments
- Defender for Endpoint integration

Compute + Data + Infrastructure

 **Secure**

Harden resources and services

Key Actions

- MCSB security policies
- Azure Policy integration
- Prioritized recommendations
- Secure score tracking
- Scope: mgmt group / sub / tenant

Policy-Driven Hardening

 **Defend**

Detect and resolve threats

Key Actions

- Security alert generation
- Kill-chain attack correlation
- Remediation guidance
- Advanced threat protection
- Just-in-time VM access

Threat Detection + Response

Coverage: Azure-native | Hybrid (Azure Arc) | Multicloud (AWS, GCP)

Knowledge check

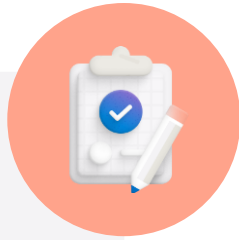


1 Which Azure capability helps enforce conditions like MFA or location before sign-in is granted?

- ☐ Role Based Conditional Access (RBAC)
- ☐ Conditional Access
- ☐ Azure Policy initiative

Summary

Learning Path 2: Describe Azure Architecture and Services



What you learned

- Regions and availability zones provide resiliency through geographic and datacenter-level fault isolation.
- Compute choice should match workload shape: VMs for control, AKS for orchestrated containers, Functions for event-driven execution.
- Storage, networking, and identity/security services form the base platform triangle for nearly every Azure solution.

End of presentation



